Information & Communication Engineering

CentraleSupélec

Communication Theory

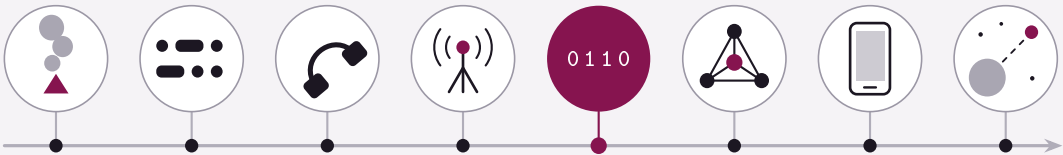
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What is communication?

> *"The imparting or exchanging of information by speaking, writing, or using some other medium"*
> — Oxford dictionary



From inter-tribal to inter-planetary

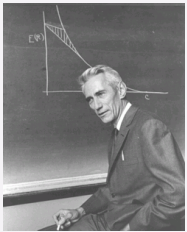
Voyager 1: one light-day away, 160 bits per second

From months to milliseconds

From a few words per minute to gigabits per second

1948: a mathematical theory of communication

> *"The fundamental problem of communication is that of reproducing at one point either exactly or approx-imately a message selected at another point."*
>
> Claude E. Shannon (1916–2001)



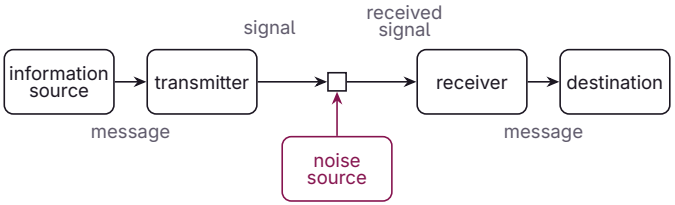
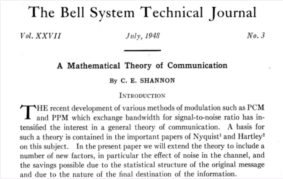


Figure 1 of the paper: a general communication system



The Bell System Technical Journal

Vol. XXVII July, 1948 No. 3

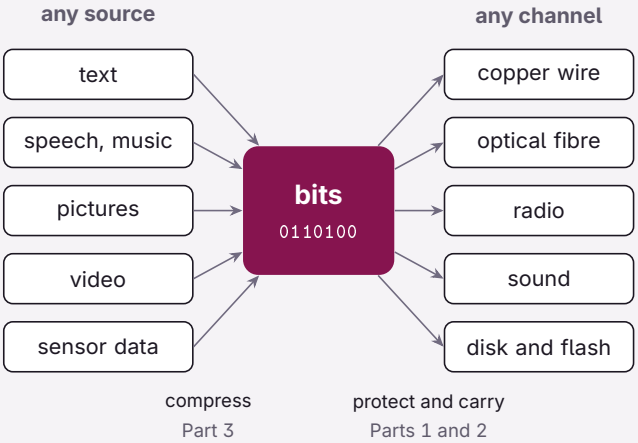
A Mathematical Theory of Communication

By C. E. SHANNON

INTRODUCTION

THE recent development of various methods of modulation such as PCM and PPM which exchange bandwidth for signal-to-noise ratio has intensified the interest in a general theory of communication. A basis for such a theory is contained in the important papers of Nyquist[1] and Hartley[2] on this subject. In the present paper we will extend the theory to include a number of new factors, in particular the effect of noise in the channel, and the savings possible due to the statistical structure of the original message and due to the nature of the final destination of the information.

Digital communication: everything goes through bits



Why a binary interface?

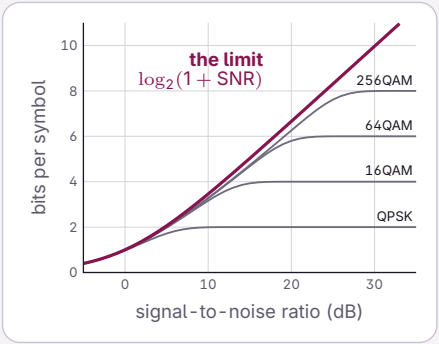
- Nothing is lost: it is optimal, according to information theory
- Source and channel can be dealt with separately
- Bits can be “cleaned up”, stored and multiplexed
- Cheap, reliable, easy to standardize

The promise: reliable communication up to a limit

- Every channel has a **capacity** C.
- Below C, coding makes the error probability as small as we want.
- Above C, nothing can.

In a band W, with power P and white Gaussian noise N_0:

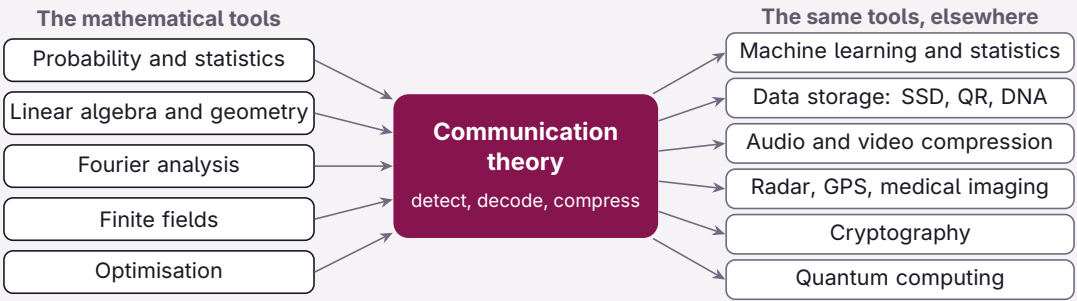
$$C = W \log_2\left(1 + \frac{P}{N_0 W}\right) \quad \text{bits per second}$$



1948: the limit. 1993–2009: codes that come close (turbo, LDPC, polar). They are now in every phone, and in the demo that follows.

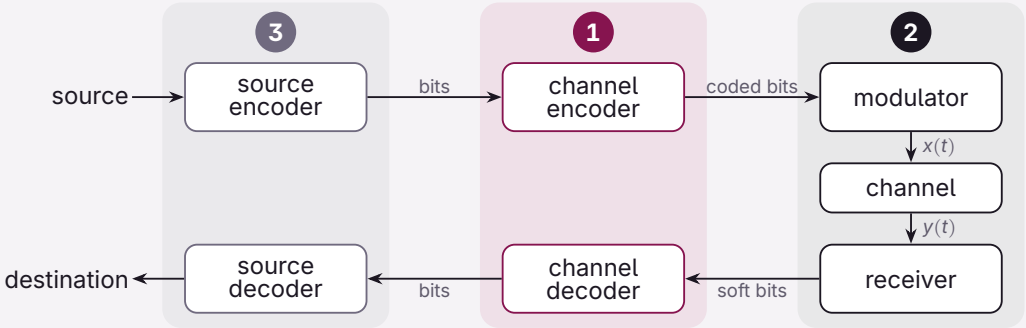
About this course

A course of **applied mathematics**: we explain, with proofs, how and why today's digital communication systems work.



What you need: probability, linear algebra, signals and systems

One chain, three parts



Source coding and limits

How few bits?
How fast at best?

Lectures 9 to 11

Channel coding

How do we protect bits against errors?

Lectures 1 to 4

Signals and receivers

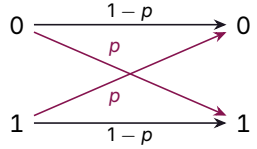
How do bits travel as waveforms?

Lectures 5 to 8

Part 1 • Channel coding

A channel makes errors or erasures. How much can a code repair, and how do we decode it?

Lecture 1	Channels and optimal decoding
Lecture 2	Linear block codes, Hamming codes
Lecture 3	Convolutional codes, Viterbi algorithm
Lecture 4	Reed–Solomon codes
TD1, TD2	Decoding by hand, step by step
TP1	Your own Hamming and Viterbi decoders
TP2	Reed–Solomon and a QR code to scratch



A channel that flips bits

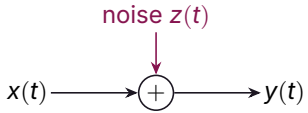


Scratched, and still readable

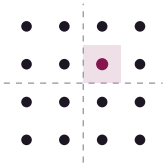
Part 2 • Signals and receivers

Bits travel as waveforms through wires, radio or air. Where do the channels of Part 1 come from?

Lecture 5	Signals as vectors, sampling theorem
Lecture 6	Modulation, pulses, Nyquist criterion
Lecture 7	Noise and the matched filter
Lecture 8	Detection and soft decisions
TD3	Pulses, noise and detection, by hand
TP3	Your own receiver for a real room with echoes



The channel adds noise

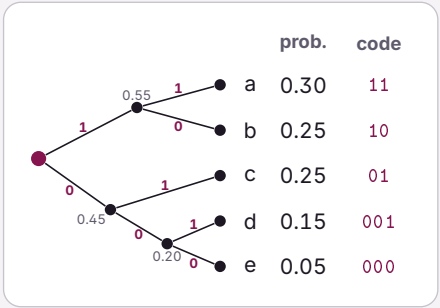


16QAM: one point carries 4 bits

Part 3 • Source coding and limits

How few bits describe a source? How many bits can a channel carry? What happens when we combine the two?

Lecture 9	Lossless compression: Huffman codes
Lecture 10	Lossy compression: quantization
Lecture 11	Capacity, source–channel separation
TD4	Huffman codes and quantizers
TP4	Send the best picture through the air



A Huffman code: frequent symbols get short codewords

The 19 sessions



11 lectures
on the blackboard, 16.5 hours

4 TDs
exercise sessions, 6 hours

4 TPs
lab sessions, 6 hours

How we work

Lectures

- On the blackboard: definitions, proofs, examples
- Lecture notes on the course website, lecture by lecture
- Take your own notes

TD

- One exam-style problem on paper in every TD
- Interactive web pages: you compute, the page checks every step
- Bring a laptop, a tablet or a phone

TP

- Python notebooks: you write the decoders and the receiver
- Real rooms, recorded with a phone
- Attendance is mandatory

Course website lecture notes, TDs with their corrections, TPs with their solutions

Evaluation

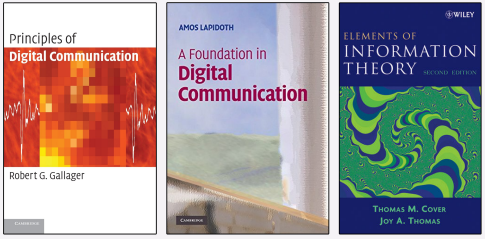
Final exam

- **2 hours**, written
- Documents are allowed
- It gives the **final score** of the course

The four TPs

- **Mandatory**; your work is handed in at the end of each session
- Evaluated as part of the “compétence”
- They do not count in the final score

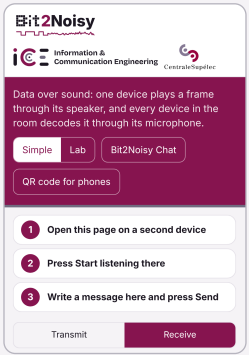
References



- R. G. Gallager, *Principles of Digital Communication*, Cambridge University Press, 2008
- A. Lapidoth, *A Foundation in Digital Communication*, Cambridge University Press, 2009
- T. M. Cover and J. A. Thomas, *Elements of Information Theory*, Wiley, 2nd edition, 2006
- The channel coding part is based on the lecture notes of G. D. Forney, available on `dspace.mit.edu`

Further reading: D. MacKay, *Information Theory, Inference, and Learning Algorithms*; J. Soni and R. Goodman, *A Mind at Play*; J. Gleick, *The Information*.

Demo: data through the air, by sound



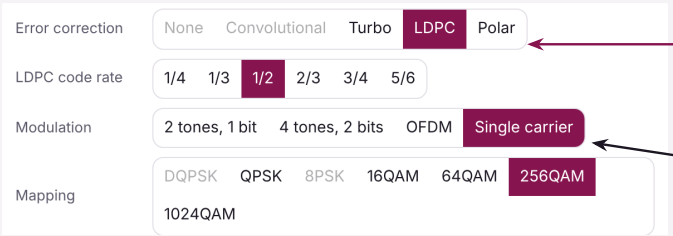
Bit2Noisy

- One device plays a frame through its loudspeaker
- Every phone in the room decodes it through its microphone
- No cable, no Wi-Fi, no Bluetooth: only sound

Open the page, choose *Receive*, press *Start listening*.

mention-ice.github.io/Bit2Noisy

What you just heard is this course



Part 1 **Error correction**
Which code, which rate, and how to decode it

Part 2 **Modulation**
Bits to sound and back, through a room with echoes

Information rate **30.3 kbit/s** = 60.6 kbit/s on the air × LDPC code of rate 1/2
Over the whole 1.1 s of sound, control part included: 176 message bits, 160 bit/s

Part 3 **Information rate**
How far is it from the limit of this room?

Now, lecture 1. The phone does not hear bits, it hears numbers. What should a receiver do with them?